

Generic Feedback on Examination for Multivariable & Vector Calculus 2MVA/2MVA(3) (25667/35172) Part B

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General comments

Part B of Multivariable & Vector Calculus (25667/35172) was taught by Dr Natalia Petrovskaya and was examined by Questions 2 and 4 on the examination paper. The examination questions were based upon all aspects of the syllabus taught in the second half of autumn term. Each question was marked out of 25. Students' hard work has been appreciated and students who made themselves confident with all educational material taught in the second half of autumn term obtained good marks in the final examination with some students obtained marks greater than 90% in both questions.

General comments about Question 2

Question 2(a): This question was based on the definition of the triple scalar product given in lectures. A common mistake in answering the question was proving the relationship between the three vectors defining the triple product without giving the definition of it first.

Question 2(b): A simple question entirely based on the definition of the unit tangent vector. Most students answered it well. Some students missed the requirement of the unit length and did not scale the tangent vector.

Question 2(c): Many students answered the question based on a wrong definition of a conservative vector field. Instead of using the definition, they mentioned some properties of the conservative field without proving those properties and lost marks because of it.

Question 2(d): Many students opted for applying the Green's Theorem in the solution and got full marks in this question. Some students made computational mistakes resulted in a wrong answer.

General comments about Question 4

Question 4(a): Part (i) of this question was entirely based on definitions and many students coped very well with it. Some students gave a wrong definition of the gradient vector in the question and they considered it as a scalar quantity. In part (ii), careful application of the chain rule was required to obtain a correct answer and some students failed to differentiate the function correctly.

Question 4(b): The question required application of the Gauss' Divergence Theorem and many students submitted a correct answer to this question. Some students lost marks because they did not use spherical coordinates correctly.

Where students typically did well

Most students did very well in Question 2(d) and showed solid understanding of how application of vector calculus theorems allows one to simplify evaluation of integrals. It was also good to see that most students understood well the topic of parametrisation in Question 2(b).

Typical areas for improvement

The Chain Rule is employed across a wide range of mathematical problems and good understanding of the Chain Rule is an important component of any mathematics-related degree. The Chain Rule should be revised by students who do not feel confident with its application.

The Gauss' Divergence Theorem is a very important topic and it is used heavily in solution to many real-life physical problems. Students lost their marks in Question 2(d) because they did not understand how to implement the Gauss' Divergence Theorem and that topic should be revised.

In any assessment, it is important that students read all of the questions carefully and use all information provided in the question. Students lost their marks because they ignored some important details in the question formulation. Also, some students did not show all steps in their solution, while the requirement of providing the full answer to the question step by step is important and cannot be ignored.